

Salem A. Bahadi

Ph.D. Candidate, Chemical Engineering

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PROFESSIONAL SUMMARY

Ph.D. in Chemical Engineering (KFUPM) with a track record in advanced nanomaterial synthesis, environmental remediation, and drilling fluid optimization, published in journals such as *Separation and Purification Technology* and *Environmental Research*. Expertise spans the engineering of multicomponent adsorbents (GO/Clay/MgFeAl-LTH) and functionalized zeolitic imidazolate frameworks (ZIFs) for H₂S scavenging and wastewater treatment. Highly skilled in linking experimental characterization—using UV-Vis, rheometry, zeta potential, XRD, FT-IR, and TGA—with computational forecasting through hybrid AI and ensemble machine learning models. Seeking to translate this foundation in nanomaterial design, adsorption thermodynamics, and predictive modeling toward next-generation environmental technologies, complex fluid systems, and sustainable process engineering.

EDUCATION

01/2023 – Present Dhahran, KSA	Ph.D. Chemical Engineering <i>King Fahd University of Petroleum and Minerals</i> Research Area: Drilling Fluid & H₂S Scavenging . CGPA: 3.821/4.0 Expected Graduation: 12/2026
08/2019 – 12/2022 Dhahran, KSA	MSc. Chemical Engineering <i>King Fahd University of Petroleum and Minerals</i> Thesis title: The formulation and characterization of novel adsorbents using metal hydroxides, graphene, and clay for the effective removal of organic pollutants from wastewater. CGPA: 3.786/4.0 .
01/2015 – 01/2019 Dhahran, KSA	B.S. Chemical Engineering <i>King Fahd University of Petroleum and Minerals</i> Second Honor. Overall G.P.A. 3.535/4.0 .
08/2017 – 12/2017 Fl, USA	Student Exchange Program <i>University of Florida</i>

AWARDS

- **1st Place**, Graduate Division – MENA Region Student Design Competition, **Digital Flow Assurance Symposium**, Doha, Qatar (2025).
- **Second Place**, **5th KFUPM Student Research Contest (SRC)** (2025), for research on advanced adsorbent materials for wastewater treatment.
- **Second Place**, **Chem-E-Car Competition**, MEPEC Conference, Bahrain – Represented KFUPM as part of Team *Sadeem*, ranking second among Middle East teams.
- **Certificate of Appreciation**, KFUPM AIChE Student Chapter, in recognition of contributions to student activities and events.
- **Certificate of Achievement**, Mentor, CITethon – CITE Conference (2025).
- **Certificate of Participation**, SPE Petroleum Economics Symposium (1.6 CEUs).

RESEARCH EXPERIENCE

01/2025 – Present	Drilling Fluid & H₂S Scavenger <i>Chemical Engineering Department, KFUPM</i> Engineered functionalized water- and oil-based drilling fluid systems for H ₂ S removal, systematically optimizing their rheological properties to ensure operational stability and efficiency.
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01/2018 – 12/2025	Wastewater Treatment <i>Chemical Engineering Department, KFUPM</i> Engineered novel multi-component adsorbents, including GO/Clay/MgFeAl-LTH and Layered Triple Hydroxide/ZIF nanocomposites, achieving highly effective removal of anionic/cationic azo dyes, phenolic compounds, and heavy metals from contaminated water.
01/2024 – 12/2024	AI & Machine Learning in Chemical Processes <i>Chemical Engineering Department, KFUPM</i> Developed and applied optimal hybrid artificial intelligence and ensemble meta-machine learning models to accurately predict the adsorptive removal of dyes and phenols from aqueous solutions, reducing experimental dependency and optimizing process design.
08/2019 – 05/2021	Crude Oil Emulsions Stabilized by Biosurfactant <i>Chemical Engineering Department, KFUPM</i> Formulated and characterized crude oil-in-water nanoemulsions stabilized by environmentally friendly biosurfactants. Investigated the complex fluid stability and successfully demonstrated pH-switchability, allowing for controlled, stimulus-responsive demulsification.

WORK EXPERIENCE

2019 – Present KFUPM, KSA	Undergraduate Research Supervisor <i>Chemical Engineering Department</i> Supervised more than ten undergraduate researchers across multiple funded projects in wastewater treatment, drilling fluid, and H ₂ S scavenging. - Guided students through experimental design, instrumentation use (Rheometer, UV-vis), and data interpretation. - Contributed to project outputs, including peer-reviewed publications.
2017 – 2022 KFUPM, KSA	Teaching Assistant <i>Chemical Engineering Department</i> Graded assignments and provided office-hour support across multiple undergraduate and graduate courses, including transport phenomena (Fluid mechanics (CHE-204), Mass transfer (CHE-304), Heat transfer (CHE-300)), Stagewise separation (CHE-306), Thermodynamics (CHE-303), Chemical engineering laboratory I (CHE-309), Kinetics, reactor design (CHE-402), Numerical methods in CHE (CHE-560), etc.
2023 – Present KFUPM, KSA	Lab Instructor <i>Chemical Engineering Department</i> Delivered hands-on instruction in Chemical Engineering Laboratory II (CHE 409) to undergraduate cohorts. - Designed and led experimental sessions covering reaction kinetics, process control, and unit operations. - Provided structured feedback on data interpretation and safety practice to groups of 16 students per semester.
2025 – Present	Peer Reviewer <i>Various Leading Scientific Journals</i> Contributed as an invited peer reviewer for leading scientific journals, reviewing more than 20 manuscripts on materials science and chemical engineering processes.
04/2025 – 05/2025 KFUPM, KSA	Mentor <i>CITE Conference, KFUPM</i> Contributed as a designated Mentor during the CITEthon competition, providing technical and structural guidance to participating teams.
05/2025 – 05/2025 Al Khobar, KSA	Technical Session Scribe <i>SPE Petroleum Economics Symposium</i> Documented technical presentations and panel discussions on petroleum economics and energy transition, preparing accurate records of key findings and industry insights.

LAB INSTRUMENTATION

FT-IR | XRD | TGA | ZetaSizer (DLS) | UV-vis | Rheometer

PUBLICATIONS

- **S.A. Bahadi**, Q.A. Drmosh, Adsorption of Anionic and Cationic Azo Dyes from Wastewater Using Novel and Effective Multicomponent Adsorbent, Sep. Purif. Technol. (2024). <https://doi.org/10.1016/j.seppur.2024.126402>.
- S.A. Onaizi, M. Alsulaimani, M.K. Al-Sakkaf, **S.A. Bahadi**, M. Mahmoud, A. Alshami, Crude oil/water nanoemulsions stabilized by biosurfactant: Stability and pH-Switchability, J. Pet. Sci. Eng. 198 (2021) 108173. <https://doi.org/10.1016/J.PETROL.2020.108173>.
- **S.A. Bahadi**, Q.A. Drmosh, Adsorptive removal of organic pollutants from aqueous solutions using novel GO/bentonite/MgFeAl-LTH nanocomposite, Environ. Res. (2024). <https://doi.org/10.1016/j.envres.2024.118218>.
- G.M. BinMakhashen, **S.A. Bahadi**, H.A. Al-Jamimi, S.A. Onaizi, Ensemble meta machine learning for predicting the adsorption of anionic and cationic dyes from aqueous solutions using Polymer/graphene/clay/MgFeAl-LTH nanocomposite, Chemosphere (2023) 140861. <https://doi.org/10.1016/J.CHEMOSPHERE.2023.140861>.
- M.K. Al-Sakkaf, I. Basfer, M. Iddrisu, **S.A. Bahadi**, M.S. Nasser, B. Abussaud, Q.A. Drmosh, S.A. Onaizi, An Up-to-Date Review on the Remediation of Dyes and Phenolic Compounds from Wastewaters Using Enzymes Immobilized on Emerging and Nanostructured Materials: Promises and Challenges, Nanomaterials 13 (2023). <https://doi.org/10.3390/nano13152152>.
- H.A. Al-Jamimi, **S.A. Bahadi**, G.M. BinMakhashen, S.A. Onaizi, Optimal hybrid artificial intelligence models for predicting the adsorptive removal of dyes and phenols from aqueous solutions using an amine-functionalized graphene oxide/layered triple hydroxide nanocomposite, J. Mol. Liq. 391 (2023) 123374. <https://doi.org/10.1016/j.molliq.2023.123374>.
- **S.A. Bahadi**, M. Iddrisu, M.K. Al, S. Mohammed, W. Ahmed, A. Amrani, U. Ahmed, U. Zahid, Q.A. Drmosh, S.A. Onaizi, Chemically versus thermally reduced graphene oxide: effects of reduction methods and reducing agents on the adsorption of phenolic compounds from wastewater, Emergent Mater. (2023). <https://doi.org/10.1007/s42247-023-00514-y>.
- A. Ibrahim, M.S. Vohra, **S.A. Bahadi**, S.A. Onaizi, M.H. Essa, T. Mohammed, Heavy metals adsorption onto graphene oxide: effect of mixed systems anresponse surface methodology modeling, Desalin. Water Treat. 266 (2022). <https://doi.org/10.5004/dwt.2022.28615>.
- **S.A. Bahadi**, Q.A. Drmosh, S.A. Onaizi, Synergism between polyethyleneimine, graphene oxide, and MgFeAl-layered triple hydroxide in removing acid red 1 dye and bisphenol A from contaminated water samples, Colloids Surf. A Physicochem. Eng. Asp. 688 (2024) 133579. <https://doi.org/10.1016/j.colsurfa.2024.133579>.
- M. Iddrisu, **S.A. Bahadi**, M.K. Al-Sakkaf, U. Ahmed, U. Zahid, Q.A. Drmosh, S.A. Onaizi, Harnessing zeolitic imidazolate framework-8 (ZIF-8) nanoparticles for enhancing H₂S scavenging capacity of waste vegetable oil-based drilling fluids, Emergent Mater. 7 (2024) 591–602. <https://doi.org/10.1007/s42247-023-00535-7>.
- **S.A. Bahadi**, M. Iddrisu, M.K. Al-Sakkaf, M.A.A. Elgzoly, Q.A. Drmosh, W.A. Al-Amrani, U. Ahmed, U. Zahid, S.A. Onaizi, Optimization of methyl orange adsorption on MgFeAl-LTH through the manipulation of solution chemistry and synthesis conditions, Emergent Mater. (2023). <https://doi.org/10.1007/s42247-023-00513-z>.
- M. Iddrisu, M.K. Al-Sakkaf, **S.A. Bahadi**, U. Zahid, Q.A. Drmosh, U. Ahmed, S.A. Onaizi, Waste to a commodity: the utilization of waste cooking oil for the formulation of oil-based drilling mud with H₂S scavenging capability bestowed by the incorporation of ZIF-67, Emergent Mater. (2023). <https://doi.org/10.1007/s42247-023-00531-x>.
- **S.A. Bahadi**, A.H. Hassan, S.A. Onaizi, Recent advances in CO₂ adsorption using aerogel materials: A review, Next Materials 7 (2025) 100388. <https://doi.org/10.1016/j.nxmte.2024.100388>.
- **S.A. Bahadi**, W.A. Al-Amrani, Q.A. Drmosh, M. Hossain, S.A. Onaizi, Adsorption of Organic and Inorganic Pollutants from Wastewater Using Various Effective Adsorbents: A Comparative Study, Water Air Soil Pollut. 236 (2025). <https://doi.org/10.1007/s11270-025-08593-9>.

- **S.A. Bahadi**, M.Q. Alkoshab, W.A. Al-Amrani, Q.A. Drmash, S.A. Onaizi, Removal of Various Dyes, Phenols, and Heavy Metals from Wastewater using Calcined and Non-Calcined Layered Triple Hydroxide/Zeolitic Imidazolate Framework Nanocomposites, Water-Energy Nexus (2026). <https://doi.org/10.1016/J.WEN.2026.06.001>.

PATENTS

- Onaizi, S.A., **Bahadi, S.A.** "Multi-nanocomposite for water decontamination." U.S. Patent Application 18/912,414 (2026).

CONFERENCES

- **Bahadi, S.**, Drmash, Q., Onaizi, S. "Effective Cationic and Anionic Dye Removal from Wastewater Using Multi-Functional Adsorbent." 21st Global Congress on Process Safety (2025).
- Elsayed, A., Murtaza, M., **Bahadi, S.**, et al. "Water-Based Drilling Fluid Containing ZIF-8@ Graphene Composite for Effective Hydrogen Sulfide Scavenging." 2025 AIChE Annual Meeting.

LANGUAGES

- Arabic (First Language)
- ENGLISH (Fluent)

REFERENCES

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